Solution Requirements (Functional & Non-functional)

Date:	29 September 2026
Team ID:	SWTID-2026-6934
Project Name:	FitSense AI – Personalized Fitness Recommendations Powered by AI
Maximum Marks:	4 Marks

1. **TEAM ID:** SWTID-2026-6934
2. **TEAM LEADER:** SAMUEL JOSHUA R & samrajan0210@gmail.com
3. **TEAM MEMBERS:**
    - ROOHITH S & roohith0529@gmail.com
    - SAMVEL M & mssamvel777@gmail.com
    - SATHISHKUMAR B & baskersathishbaskersathish@gmail.com

Functional Requirements: Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration via email and password • Secure password hashing via bcrypt.js
FR-2	User Authentication	• Secure login via credentials • JWT token generation and validation middleware
FR-3	Contact Management	• Add new contact (Name, Email, Phone, Notes, Tags) • View contact list & details • Update existing contact information • Delete contact records
FR-4	Search & Categorization	• Tag-based contact categorization • Real-time dynamic search by name, email, or tags

Non-functional Requirements: Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean RESTful JSON APIs structured for effortless client integration.
NFR-2	Security	Strict endpoint protection using token validation headers and encrypted passwords.
NFR-3	Reliability	Centralized error-handling middleware ensuring graceful JSON error responses.
NFR-4	Performance	Optimized database querying via Mongoose schema indexes.
NFR-5	Availability	Stable server execution supporting continuous local or cloud uptime.
NFR-6	Scalability	Modular MVC separation allowing future feature additions seamlessly.